Data purity, not model capability, can dominate apparent AlphaFold3 performance on phosphorylation benchmarks

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November 13, 2026

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Abstract

Motivation: AlphaFold3 (AF3) accepts explicit post-translational modifications, which is widely taken to imply that it can model the conformational consequences of phosphorylation. Testing that implication requires a pair of experimental structures that are genuinely comparable, yet we could not find a published phosphorylation benchmark that audits this, so complex-composition effects can be mistaken for phosphorylation effects.

Results: We assembled a phosphorylation structure benchmark and subjected every pair to a five-criterion biological purity gate (P1–P5). Of 24 screened candidates, 6 were dropped for lacking a verified phospho-residue, leaving 18 curated entries: 16 are two-state pairs and 15 have AF3 models for both states, of which 9 pass all five criteria. On these 15 pairs the correlation between AF3-predicted and experimentally observed mean Cα displacement (one value per protein) is not significant ($r = +0.11$, 95% CI $[-0.34, +0.57]$, $n = 15$), while the predicted displacement is nearly constant (0.52 ± 0.36 Å) against an experimental span of 0.25–2.51 Å. Both AF3 internal confidence signals fare no better: the association between site pLDDT and predictive agreement ($r = +0.48$) and that between ΔPAE and predictive agreement ($r = +0.60$) are each carried by a single leverage point and collapse when it is removed ($r = +0.21$ and $r = +0.14$). Restricting to purely paired structures changes the headline statistic: pairing purity, not model capability, can govern the apparent result ($r = +0.22$ for the 9 pure pairs versus $r = -0.24$ for the 6 that fail the composition check; both intervals include zero and a permutation test on their difference gives $p = 0.40$). A second, sequence-only ESMFold line, run with a phosphomimetic substitution, fails to track the cohort-level response in the same way ($r = -0.36$), although like AF3 it retains per-protein magnitude correlation.

Availability and Implementation: Benchmark data, analysis code and frozen intermediate results are available at https://github.com/tjogzt/ptm-phospho-af3-benchmark (archived at Zenodo: https://doi.org/10.5281/zenodo.24087962); implemented in Python 3.12.
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Supplementary information: Supplementary figures and tables are included within this preprint.

Keywords: AlphaFold3, phosphorylation, conformational change, benchmark, data purity, prediction confidence

1 Introduction

Phosphorylation is the most prevalent regulatory post-translational modification (PTM). By altering local electrostatics, hydrogen bonding and backbone flexibility, a single phosphate group can propagate structural rearrangements far from the modified residue, switching a protein between functional states (Johnson and Lewis, 2001). Reconstructing the structural consequences of phosphorylation is therefore central to mechanistic interpretation of signalling, and it is precisely the kind of "instance-specific" question that structure-prediction models are now expected to answer.

AlphaFold3 (AF3) marked a qualitative change in this expectation: unlike earlier protein-only predictors, it accepts explicit chemical components and modified residues as input, including phosphoserine (SEP), phosphothreonine (TPO) and phosphotyrosine (PTR) (Abramson et al., 2024). Because the modification is part of the input, it seems natural to read a difference between an unmodified and a modified AF3 prediction as the model's view of the phosphorylation-induced structural change. Several studies have begun to use AF3 in exactly this "mechanistic" mode (Nam, 2026; Ramasamy et al., 2026), and AF3's per-residue confidence (pLDDT) and predicted aligned error (PAE) are frequently quoted as reassurance that the predicted change is trustworthy (Carugo, 2023; Akdel et al., 2022).

Three gaps complicate this practice. First, benchmarking structural responses to phosphorylation requires a pair of experimental structures that differ *only* in the modification. In the Protein Data Bank (PDB), however, a phospho form and a non-phospho form of the same protein are usually crystallised as different constructs: different oligomeric states, different bound ligands, different domain boundaries, sometimes different partners. Any of these covariates induces large conformational differences that have nothing to do with the phosphate, yet we could not find a published phosphorylation-pairing benchmark that audits whether the paired structures are genuinely comparable. Second, the metrics used to summarise agreement (typically a mean displacement magnitude or an amplitude ratio) are dominated by a handful of disordered, highly mobile residues, so that a large apparent "under-prediction" can be a measurement artefact. Third, pLDDT and PAE measure *self-consistency* of a single prediction, not agreement with a reference; using them as proxies for correctness of a PTM-induced change is a category error unless it is tested.

This work reports a conservative benchmark that addresses these gaps and, in doing so, returns a negative result. Our contributions are threefold.

1. **A purity audit with biological criteria.** We formalise a five-criterion gate (P1–P5) covering identity, presence of a genuine modified residue, two-state coverage, overlap about the target site, and chain-composition consistency, and we show that a substantial fraction of apparently usable phosphorylation pairs fail at least one criterion.

2. **A negative result on a cleaned cohort.** On the pairs that survive the gate we find no evidence that AF3's predicted phosphorylation-induced displacements track the experimentally observed ones, we show that AF3's predicted displacement is close to a constant, and we find that a second, sequence-only predictor (ESMFold) behaves the same way.

3. **Two falsified confidence hypotheses.** We test whether site pLDDT or ΔPAE predicts agreement, and show that both apparent associations are produced by a single leverage point and disappear when it is removed.

We do not claim that AF3 is incapable of modelling phosphorylation effects; we claim that the widely used benchmark protocol cannot currently demonstrate that it does, and we give concrete, low-cost practices that avoid the two failure modes we identify.

2 Background and related work

AF3 and explicit modified residues. AF3 takes an explicit list of residues and ligands, so a phospho form can be specified directly (Abramson et al., 2024). This capability is orthogonal to accuracy on modification-induced change, yet the two are often conflated. Prior evaluations of AF3 on modified or non-protein components (Nam, 2026) focus on local geometry around the modified residue rather than on global conformational responses, and an assessment of the relation between phosphorylation, AF3 models and conformational variability (Ramasamy et al., 2026) concludes that it remains unclear whether the models capture PTM-driven conformational change.

Structural-change benchmarks. Comparing two states of a protein is a mature problem: one superposes a common core (e.g. Kabsch (Kabsch, 1976)) and measures per-residue displacement, or decomposes the motion into rigid-block (hinge) components (Hayward and Berendsen, 1998). The choice of core and the treatment of disordered regions strongly affect the reported magnitude, which is why amplitude summaries must be interpreted against a noise model.

Confidence as a proxy for correctness. pLDDT and PAE are estimates of local and pairwise geometric confidence within a single prediction (Abramson et al., 2024). Their use as proxies for agreement with an independent reference is common but rarely validated on a paired structural-change benchmark; when it has been examined, the relationship is weak or confounded (Carugo, 2023; Akdel et al., 2022).

3 Materials and methods

3.1 Benchmark construction

Candidate pairs were collected from the PDB as pairs of structures of the same UniProt entry in which exactly one state carries a phosphorylated residue, and were labelled with an effect type (*local rearrangement*, *allosteric*, or *disorder-to-order*). After curation the benchmark contains 18 entries, of which 16 are two-state pairs; AF3 models for both states were obtained for 15 of them, and these 15 form the *primary cohort* (Table 1). All downstream analysis uses a single frozen manifest (`manifest_frozen.json`) that binds each protein to its UniProt accession, phosphorylated site(s) in UniProt coordinates, and its phospho/unphospho PDB identifiers and chains, so that every number in this paper is traceable to one immutable table.

3.2 Pairing-purity gate (P1–P5)

Each candidate pair is adjudicated by five biological criteria, evaluated independently for the phospho and non-phospho structures.

P1 (identity). Both structures map to the same UniProt entry at high sequence identity.

P2 (genuine modification). The phospho structure contains at least one modified residue (SEP, TPO or PTR) recorded in the PDB MODRES records. P2 constrains only the presence of the

modification; it does not require the non-phospho state to be wild type, and engineered substitutions carried by either state are recorded as caveats (Sec. 3.2) rather than treated as disqualifying.
P3 (two-state coverage). Each state covers at least 50% of the UniProt sequence.
P4 (overlap about the site). The two states share at least 20 common residues and the shared region includes the target site.
P5 (composition consistency). The two states have the same chain composition, so that differences in oligomeric state or in bound partners cannot be mistaken for a phosphorylation effect. A pair that fails P1 (identity) or P2 (a genuine, MODRES-recorded phospho-residue) is discarded outright. P3, P4 and P5 are recorded as quality flags: the primary cohort comprises every pair that passes P1–P2 and for which models exist in both states (15 pairs, Table 1), and each cohort-level result is also reported after removing the pairs that fail P3 (Supplementary Table S1) or P5 (Sec. 4.1, Fig. 1b). Within the primary cohort, 9 pairs pass all five criteria (the *pure* cohort) and 6 carry at least one flag: all six fail P5, and two of these (AMPKα2, TBK1) also fail P3. A further pair passes the five criteria but has no model and therefore lies outside the cohort altogether. P5 violations are the most common and the most easily overlooked source of confounding, which is why the flag is displayed rather than silently acted upon: the phospho and non-phospho entries differ in chain count or partners, exactly the covariate that a naive amplitude comparison would attribute to phosphorylation.
What "pure" does and does not mean. P5 asks only that the two states share the same chain composition. It is deliberately blind to bound ligands, engineered substitutions and additional modification sites, so "pure" must be read as *chain-composition-matched*, not as "identical apart from the phosphate". Auditing the nine pure pairs (`audit_confounds.py`; `pair_confound_audit.json` in the archived repository) shows how much heterogeneity remains: seven of the nine differ in bound ligands between the two states, and only two of the sixteen paired entries (ERK2, p38α) are composition-matched in the stricter sense of sharing the same ligands and carrying no substitutions. Two of the nine pure pairs do carry engineered substitutions: the AURKA non-phospho state carries T287D, one residue from the phosphosite T288, and PAK1 carries K299R in both states. Outside the pure stratum, the IKBKB non-phospho reference (4E3C, an entry that fails P5) is an S177E/S181E phosphomimetic double mutant at exactly the two sites under study. Three phospho states additionally carry experimental phosphorylation sites that AF3 was not asked to model (CDK7 pS164; IKBKB pS257; and AMPKα2 pS108, which lies on the β subunit rather than on the target chain), and AF3 was run on the apo full-length sequence against ligand-bearing crystals. These caveats do not rescue the negative result, and they do not act in one direction: the phosphomimetic and engineered substitutions sit in the non-phospho references and therefore *remove* part of the difference the model would have to reproduce, whereas any residual covariate mismatch adds variance that biases a correlation toward zero and could manufacture a negative result. The check that answers this is the pure stratum itself, on which the negative result holds with the same sign and a wider interval ($r = +0.22$, $n = 9$, spanning zero; Sec. 4.1); these caveats therefore bound what "pure" means rather than weakening the conclusion.

3.3 Prediction protocols

AlphaFold3. For each protein we submitted the full-length UniProt sequence with the approved site(s) specified as modified residues using the corresponding CCD codes, and collected 15 models per state (five diffusion samples across three random seeds). The displacement and amplitude statistics aggregate all 15 models; the single-model exceptions are stated where they arise — the pLDDT, geometric-decomposition and residue-spectrum analyses use the first diffusion sample of the first seed, and the ΔPAE analysis uses model 0 across seeds (Secs. 4.5 and 4.6). The seed axis is treated as sampling repetition, not biological replication.

ESMFold (comparison line). As an independent, sequence-only predictor we folded the same full-length sequences with ESMFold (Lin et al., 2023). ESMFold does not parse CCD modified residues, so the phospho state was represented by a phosphomimetic substitution (pSer/pThr→Asp, pTyr→Glu); this limitation is stated explicitly and the comparison is therefore qualitative.

3.4 Displacement metrics

All comparisons share one coordinate protocol. The non-phospho experimental structure defines the reference frame; the phospho experimental structure is superposed onto it over common Cα atoms by the Kabsch algorithm (Kabsch, 1976), giving the experimental displacement field Δ_{exp}. Predicted fields are obtained by superposing each predicted state onto the common experimental non-phospho reference frame with the Kabsch algorithm, independently for the two states, so that each state's arbitrary internal frame is removed while both remain expressed in one common frame, giving Δ_{pred}. Because the two predicted states are independent samples of unrelated conformers, only two rigid-body conventions are physically defined — independent superposition onto the experimental references (used throughout) and direct superposition of the two predicted states onto each other; both yield the same cohort-level result (Supplementary Table S1). From these fields we compute: (i) the per-residue displacement magnitude; (ii) the amplitude ratio, $|\Delta_{\mathrm{pred}}|/|\Delta_{\mathrm{exp}}|$; (iii) the direction cosine between Δ_{pred} and Δ_{exp}; (iv) the Pearson correlation r of the per-residue magnitude profiles; (v) a hinge index from a rigid-block superposition; and (vi) a site window of ± 10 residues around each modified site. Modified residues (SEP/TPO/PTR) are retained throughout; omitting them, as a naive HETATM filter does, silently removes the residue of interest from both sides of the comparison.

3.5 Statistics

The independent unit is the *protein*, not the model: the 15 AF3 models of one protein share a single experimental reference and are not independent observations. Cohort correlations are therefore computed over proteins, and bootstrap confidence intervals resample proteins, not models. Sensitivity to single points is assessed by leave-one-out and leverage diagnostics. Correlations are reported as Pearson r with Spearman ρ where a monotone alternative is relevant; no correction for multiplicity is applied to the exploratory correlations, which are labelled as such.

3.6 Same-state noise floor

To decide whether experimentally observed displacements are distinguishable from noise, we computed two *same-state* controls: (i) an experimental floor, the Cα dispersion after superposing crystallographically independent copies (NCS) of the same chain in the same state; and (ii) a predictor floor, the dispersion among the 15 AF3 models of the same state. Because AF3 predicts full-length sequences whereas experimental structures cover only the ordered core, the predictor floor is evaluated on the residues actually resolved in the experiment, so that unmodelled termini do not inflate it.

4 Results

4.1 The benchmark shrinks under a purity gate

Of 24 screened candidates, 6 were dropped for lacking a MODRES-recorded phospho-residue, leaving 18 curated entries. Of these, 16 are two-state pairs and 15 have AF3 models for both states (Fig. 1a). Applying P1–P5 leaves 9 pure pairs; the remaining 6 all fail P5, because the phospho and non-phospho entries differ in chain count or partners — exactly the covariate that a naive amplitude comparison

Protein	UniProt	Effect	PTM	Site(s)	Unphos PDB	Phos PDB	P1	P2	P3	P4	P5	Verdict	Pure	In cohort	Note
CDK2	P24941	local	pT	Thr160 (pT)	2C5N	1H1R	Y	Y	Y	Y	Y	PASS	Y	Y	
MAPK1	P28482	local	pT/pY	Thr185/Tyr187 (pT/pY)	1ERK	2ERK	Y	Y	Y	Y	Y	PASS	Y	Y	
AURKA	O14965	local	pT	Thr288 (pT)	4J8M	5DNR	Y	Y	Y	Y	Y	PASS	Y	Y	
IRF3	Q14653	allost.	pS	Ser386/Ser402 (pS)	1QWT	3A77	Y	Y	Y	Y	N	WARN	N	Y	
MAPK14	Q16539	local	pT/pY	Thr180/Tyr182 (pT/pY)	1WFC	3PY3	Y	Y	Y	Y	Y	PASS	Y	Y	
CDK7	P50613	local	pT	Thr170 (pT)	8P79	8PYR	Y	Y	Y	Y	N	WARN	N	Y	
SRC	P12931	allost.	pY	Tyr419 (pY)	7NG7	9NS0	Y	Y	Y	Y	Y	PASS	Y	Y	
PRKAA2	P54646	allost.	pT	Thr172 (pT)	2H6D	6B1U	Y	Y	N	Y	N	FAIL	N	Y	
PRKACA	P17612	allost.	pT	Thr198 (pT)	–	6QJ7	n/a	n/a	n/a	n/a	n/a	n/a	n/a	N	single state
POLR2A	P24928	disorder	pS	Ser1614/Ser1621 (pS)	–	4JXT	n/a	n/a	n/a	n/a	n/a	n/a	n/a	N	single state
PYGM	P00489	disorder	pS	Ser15 (pS)	2GJ4	1LWO	Y	Y	Y	Y	Y	PASS	Y	Y	
CDK1	P06493	local	pT	Thr161 (pT)	4YC3	9SKQ	Y	Y	Y	Y	N	WARN	N	Y	
MAPK3	P27361	local	pT/pY	Thr202/Tyr204 (pT/pY)	4QTB	2ZOQ	Y	Y	Y	Y	Y	PASS	Y	N	AF3 models missing
GSK3B	P49841	local	pY	Tyr216 (pY)	1J1B	6Y9S	Y	Y	Y	Y	Y	PASS	Y	Y	
PAK1	Q13153	local	pT	Thr423 (pT)	1YHW	3FXZ	Y	Y	Y	Y	Y	PASS	Y	Y	
AKT1	P31749	local	pT	Thr308 (pT)	3O96	4GV1	Y	Y	Y	Y	Y	PASS	Y	Y	
TBK1	Q9UHD2	local	pS	Ser172 (pS)	4IM0	4EUU	Y	Y	N	Y	N	FAIL	N	Y	
IKBKB	O14920	local	pS	Ser177/Ser181 (pS)	4E3C	4KIK	Y	Y	Y	Y	N	WARN	N	Y	

Table 1: Benchmark cohort and pairing-purity gate (P1–P5). P1 identity, P2 real phospho-residue (MODRES), P3 coverage $\geq$0.5 in both states, P4 $\geq$20-residue overlap covering the target site, P5 identical chain composition. "n/a" marks single-state entries that have no unphosphorylated counterpart, so the gate does not apply. The Pure column reports whether P1–P5 all pass; In cohort additionally requires the models to be available. Note gives the reason an entry lies outside the primary cohort. Site numbers follow UniProt (canonical) numbering; the MODRES records of the corresponding PDB entries use PDB numbering, which can be offset (e.g. 6QJ7 MODRES TPO 197 is UniProt Thr198, and 4JXT MODRES SEP 1621/1628 are UniProt Ser1614/Ser1621).

would attribute to phosphorylation — and two of these (AMPKα2, TBK1) additionally fail the P3 coverage criterion.

This is not a cosmetic filter. Stratifying the same frozen per-protein results by purity changes the apparent behaviour of the model (Fig. 1b): the median amplitude ratio is 0.46 for pure pairs but 0.30 for impure pairs, and the magnitude correlation is $r = +0.22$ ($n = 9$, 95% CI $[-0.39, +0.91]$) for pure pairs but $r = -0.24$ ($n = 6$, 95% CI $[-1.00, +0.84]$) for impure ones. Both stratum estimates are individually indistinguishable from zero at this sample size, and a permutation test on their difference is not significant ($p = 0.40$; Supplementary Table S1); the stratification is therefore evidence that the pooled mixture is unstable, not a demonstrated contrast between two populations. Any headline computed on the uncleaned cohort is a mixture of model behaviour and pairing confounding. We report all subsequent statistics on the full 15-pair cohort while flagging the pure subset, and we avoid any claim whose sign depends on the mixture; removing the two P3-failing pairs leaves the cohort correlation essentially unchanged ($r = +0.21$, $n = 13$; Supplementary Table S1).

4.2 No evidence that AF3 tracks the phosphorylation-induced displacement

Over the 15-pair cohort the per-protein mean predicted displacement is essentially uncorrelated with the experimental displacement: Pearson $r = +0.11$ (95% CI $[-0.34, +0.57]$), slope $+0.053$ Å/Å, Spearman $\rho = +0.15$ ($p = 0.59$) (Fig. 2a). The confidence interval is wide, as $n = 15$ is small, but it excludes the strong positive association that "AF3 models phosphorylation" would require. The predicted displacement is also close to constant: 0.52 ± 0.36 Å across all proteins, against an experimental span of 0.25–2.51 Å (median 1.50 Å). The model's output moves by roughly the same small amount regardless of whether the true response is negligible (PAK1, 0.25 Å) or large (Src, 2.51 Å). "Near-constant" describes this cohort-mean behaviour rather than a reproducible per-protein value: within a single protein the AF3 samples of one state are themselves unstable (median within-protein coefficient of variation of the mean displacement 0.33; 8 of 15 proteins exceed 0.3), and for CDK2 the per-sample mean ranges from 0.22 to 2.02 Å.

Leave-one-out analysis shows the coefficient is not driven by a single outlier in the sense of an

Protein	UniProt	n_{common}	exp disp (Å)	AF3 disp (Å)	AF3 ratio	AF3 cos	AF3 r	ESM disp (Å)	ESM ratio	ESM cos	ESM r	Pure
AKT1	P31749	262	2.05	0.81	0.39	-0.40	+0.73	0.08	0.04	+0.55	+0.68	Y
PRKAA2	P54646	255	2.22	0.20	0.09	+0.02	+0.10	0.04	0.02	+0.18	+0.77	N
AURKA	O14965	263	2.44	0.29	0.12	+0.34	+0.85	0.04	0.02	+0.20	+0.68	Y
CDK1	P06493	283	1.44	0.71	0.49	+0.43	+0.62	0.06	0.04	+0.41	+0.39	N
CDK2	P24941	296	0.78	0.44	0.56	+0.26	+0.37	0.10	0.13	+0.17	+0.63	Y
CDK7	P50613	259	0.53	0.18	0.34	+0.22	+0.43	0.07	0.12	-0.72	+0.10	N
MAPK1	P28482	352	1.64	1.06	0.65	+0.66	+0.61	0.07	0.04	-0.01	+0.55	Y
GSK3B	P49841	344	0.63	0.15	0.24	+0.17	+0.45	1.53	2.43	+0.14	+0.13	Y
PYGM	P00489	803	0.94	0.43	0.46	-0.08	+0.34	0.61	0.65	+0.73	+0.99	Y
IKBKB	O14920	617	1.96	0.53	0.27	+0.17	+0.38	0.05	0.02	+0.15	+0.40	N
IRF3	Q14653	232	0.33	0.73	2.22	+0.06	+0.38	0.61	1.85	+0.31	+0.33	N
MAPK14	Q16539	338	1.57	1.39	0.89	+0.70	+0.96	0.06	0.04	+0.38	+0.52	Y
PAK1	Q13153	291	0.25	0.24	0.95	–	+0.26	0.03	0.12	–	+0.00	Y
TBK1	Q9UHD2	279	1.50	0.40	0.27	+0.56	+0.40	0.05	0.03	+0.07	+0.67	N
SRC	P12931	274	2.51	0.25	0.10	-0.07	+0.03	0.40	0.16	+0.09	+0.72	Y

Table 2: Per-protein comparison of experimental versus predicted phospho-induced Cα displacement. Displacements are mean per-residue magnitudes (Å). Ratio = predicted/experimental. cos = mean direction cosine over residues moving >1 Å. r = Pearson r of the per-residue magnitude profiles.

extreme r; the largest change is the removal of Src, which raises r to +0.22, still far from significant. The per-protein picture is heterogeneous rather than uniformly null: a few proteins show high magnitude correlation (p38, $r = +0.96$; AURKA, $r = +0.85$), but several show none (Src, $r = +0.03$) even though their experimental displacements are among the largest. Aggregated over the cohort, this heterogeneity averages out to noise.

4.3 A second, sequence-only predictor fails in the same way

The non-response could in principle be specific to AF3's structure module, which consumes an MSA. We therefore repeated the identical coordinate protocol with ESMFold, a single-sequence language-model predictor, using a phosphomimetic substitution to represent the phospho state (Sec. 3.3). The outcome is the same and, quantitatively, even flatter: the cohort-mean predicted displacement is 0.25 ± 0.41 Å (against 0.52 ± 0.36 Å for AF3) and shows no association with the experimental displacement ($r = -0.36$, 95% CI $[-0.72, +0.31]$; Spearman $\rho = -0.24$, $p = 0.40$) (Table 2, Fig. 2a). For most proteins the phosphomimetic substitution moves the predicted structure by less than 0.1 Å, i.e. essentially not at all, so the near-constant prediction is not unique to AF3. Leave-one-out again exposes the small cohort: removing the lowest- displacement protein (PAK1) moves the correlation to $r = -0.47$, while no single removal produces a significant positive association. ESMFold, like AF3, does retain local shape agreement — its mean per-protein magnitude correlation is +0.50, essentially the same as AF3's +0.46, and its mean direction cosine is +0.19 against AF3's +0.22 (Table 2, Fig. 2b) — while under-predicting amplitude about an order of magnitude more severely (median predicted to experimental ratio 0.04, against 0.39 for AF3). The shared failure is thus amplitude and cohort-level tracking rather than local shape, and it appears across two predictors, two input modalities (MSA-based and sequence-only) and two ways of representing the phosphate, which argues that it is a property of the benchmark question rather than of one model.

4.4 The predicted signal sits at the same-state noise floor

The negative result is only meaningful if the experimental signal exceeds noise. At cohort level it does, and per protein the statement must be narrower. The same-state experimental floor has a median of 0.335 Å and a maximum of 1.309 Å across the seven proteins with crystallographically independent copies (the other eight pairs have a single experimental structure each, so no experimental floor can be estimated for them), whereas the phosphorylation-induced experimental displacement has a median

of 1.50 Å and a span of 0.25–2.51 Å (Fig. 3). Each of those seven proteins exceeds its own floor — marginally for IRF3 (0.33 vs 0.29 Å) and GSK3B (0.63 vs 0.50 Å) — but the largest floor (1.309 Å, IKBKB) exceeds several cohort signals (CDK7 0.53, GSK3B 0.63, CDK2 0.78, PYGM 0.94 Å) and is not usable as a cohort-wide threshold. PAK1, which has no replicate structure, has the smallest experimental displacement in the cohort (0.25 Å), below the median floor, and cannot be checked against its own noise level at all. The claim the data support is therefore per protein rather than cohort-wide: wherever a floor is measurable the phospho signal exceeds it, so the absence of predicted tracking is not an artefact of a reference noisier than the effect. The predictor's own same-state floor (median 0.389 Å, maximum 3.03 Å for AKT1, about four times that protein's own predicted displacement of 0.81 Å, and available for all 15 pairs because each has 15 models) is of the same order as its putative phospho signal (0.52 Å); AF3's between-state displacement is therefore close to its own sampling variability.

4.5 Two confidence-based hypotheses are refuted

If the model's internal confidence flagged where it was right, a user could at least triage predictions. We tested two natural hypotheses on the 15-pair cohort.

Site pLDDT. The correlation between the unphosphorylated site pLDDT and the amplitude ratio is $r = +0.48$ (Fig. 4). The relation is carried almost entirely by IRF3, whose amplitude ratio (2.22) and site pLDDT (89.9) are both extreme: removing this single protein drops the correlation to $r = +0.21$ ($n = 14$). A companion analysis using the site-minus-global pLDDT contrast ($r = +0.43$) behaves the same way.

ΔPAE. PAE is the more natural signal for a *pairwise* question such as "does the domain carrying the site move relative to the rest?". The correlation between ΔPAE (site→rest) and the amplitude ratio is $r = +0.60$ (Fig. 5), the strongest association we observe. It is again an IRF3 effect: removing IRF3 drops it to $r = +0.14$ ($n = 14$), and the analogous statistic for global ΔPAE falls from $r = +0.49$ to $r = +0.06$.

Both confidence signals are thus indistinguishable from single-point artefacts on this benchmark. Their apparent strength comes from a small cohort containing one protein whose prediction and reference happen to disagree in both magnitude and confidence in the same direction.

4.6 Exploratory: what does the model respond to?

We therefore asked whether AF3 under-responds most where the true motion is largest. Across the 15 pairs the amplitude ratio is inversely related to the overall deformation magnitude ($r = -0.54$, $p = 0.040$ against the full-structure RMSD; $r = -0.58$, $p = 0.023$ against mean displacement) and to the block-wise deformation ($r = -0.53$, $p = 0.041$); the relation with the hinge index is weaker and not significant ($r = -0.38$, $p = 0.16$). A residue-level view agrees in sign: binning all 5,148 common residues across the 15 proteins by experimental displacement, the median predicted/experimental ratio falls steeply from 0.50 in the smallest-displacement bin to 0.07 in the largest, monotonically across the six bins (0.50, 0.31, 0.27, 0.22, 0.17, 0.07); AF3's response saturates while the true response grows (Fig. 6). This analysis uses the first AF3 model of each state rather than a multi-model average. A distance decomposition is weaker still — the bin ratios vary non-monotonically (0.15–0.31) with no trend, and the lowest ratio occurs in the immediate 0–5-residue window where the experimental displacement is largest, not at long range — so we do *not* support a hinge-geometry explanation. Because these stratifications are exploratory and uncorrected for multiplicity, we present them as hypotheses, not conclusions; in particular, the hinge comparison and any distance dependence are not significant and we do not claim that hinge motion is harder to predict.

5 Discussion

What the benchmark does and does not show. Our central empirical claim is narrow: on a 15-pair phosphorylation structure benchmark, AF3's predicted conformational response is not related to the observed one, and the two most commonly cited internal confidence signals do not identify where it succeeds. We deliberately do *not* claim that AF3 cannot model phosphorylation. The cohort is small; the metric is a single, rigid-body-based displacement field; and "no detectable correlation at $n = 15$" is weaker than "no correlation". The current, commonly used protocol cannot demonstrate the capability it is assumed to have.

Why the result is easy to miss. Two artefacts work in the same direction. First, pairing impurity inflates apparent agreement: phospho and non-phospho entries that differ in oligomeric state or partner differ in structure for reasons unrelated to the phosphate, and pooling such pairs changes the sign of the cohort correlation (Sec. 4.1, Fig. 1b). Second, amplitude summaries are dominated by a few highly mobile residues, so a large apparent "under-prediction" can reflect disorder rather than a mechanical failure. Both artefacts are avoidable at essentially no cost: apply a composition check before pairing, and report displacement statistics against a same-state noise floor.

Confidence effects reduce to a single point. Our two refutations share a mechanism: IRF3 is extreme in both the predictor's confidence and in its disagreement, and in a cohort of 15 one such protein can manufacture a correlation of 0.5–0.6. This is a warning about study design as much as about AF3: on small structural benchmarks, leverage diagnostics are not optional.

Practical guidance. (i) Audit pair purity before computing any agreement statistic; a composition check (P5) removes the largest confounder. (ii) Never drop modified residues from the comparison; they are the object of study. (iii) Calibrate measured displacements against same-state replicates; an effect smaller than the same-state floor is not interpretable. (iv) Do not use pLDDT or PAE as evidence that a PTM-induced change is correct; test that association explicitly and check its leverage.

Limitations. The cohort is modest ($n = 15$, 9 pure) and kinase-dominated: by curated effect type it comprises 11 local rearrangements, 3 allosteric effects and 1 disorder-to-order transition, so the non-local strata hold too few proteins to support separate statistics and we do not report them as such. ESMFold is included as a second, sequence-only line; because it cannot represent CCD modified residues its phospho state is phosphomimetic, so that line is qualitative, yet it agrees with AF3 in showing no tracking ($r = -0.36$). No molecular-dynamics reference is used. These limitations bound the generality of the negative result; they do not affect the two methodological findings, which are properties of the protocol.

6 Conclusion

On a phosphorylation structure benchmark cleaned by biological purity criteria, AF3's predicted conformational response does not track the experimental one, its predicted displacement is near-constant, and its internal confidence signals fail to identify where it is right once single-point leverage is removed. The broader lesson is methodological: on paired structural-change benchmarks, pairing purity and same-state noise calibration, not model capability, can dominate the apparent result, and both are cheap to enforce.

Availability and implementation

The frozen benchmark manifest, analysis scripts (Python 3.12), and all intermediate results are archived at https://github.com/tjogzt/ptm-phospho-af3-benchmark (Zenodo DOI: https://doi.org/10.5281/zenodo.24087962). The analysis is deterministic given the frozen manifest; random seeds

are fixed. AF3 predictions were generated with the AlphaFold Server (alphafoldserver.com); ESMFold inference was run locally with `transformers` on macOS (Apple Silicon, MPS backend). Codes are released under the MIT licence.

Funding

This work was supported by the National Natural Science Foundation of China (Grant Nos. 82403759 to C.X. and 82403616 to Y.W.).

Conflict of interest

None declared.

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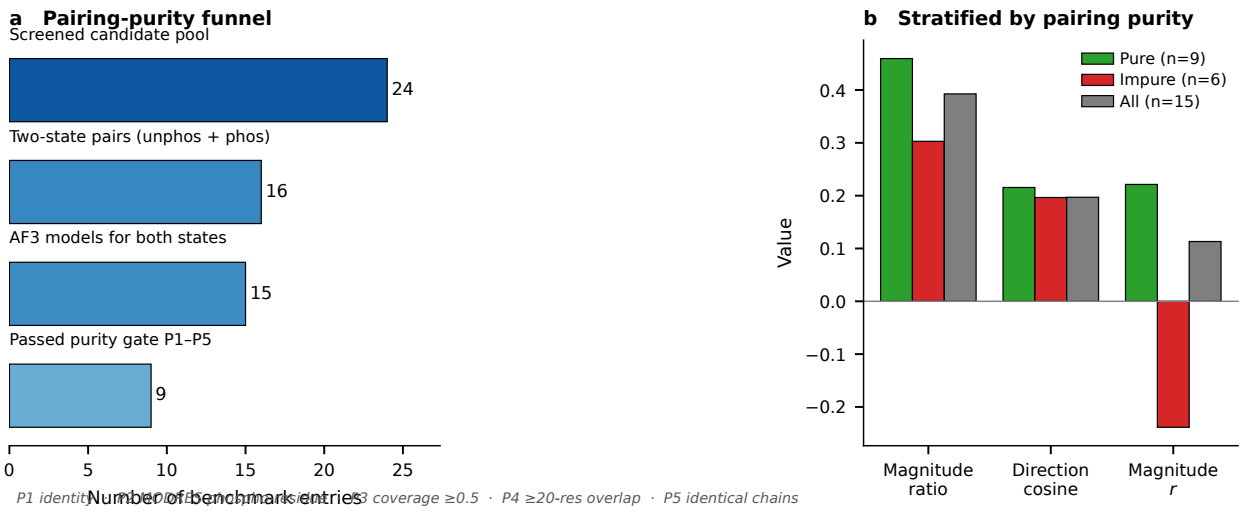


Figure 1: **Pairing purity shapes the benchmark and the conclusion.** (a) Funnel from the screened candidate pool through the two-state and AF3-availability filters to the pairs passing the P1–P5 purity gate. (b) The same frozen per-protein results stratified by pairing purity, showing the median amplitude ratio, median direction cosine and magnitude correlation for pure ($n = 9$), impure ($n = 6$) and all ($n = 15$) pairs. Purity here means chain-composition matching (P5), not ligand matching: the nine pure pairs are not identical apart from the phosphate, and two of them carry engineered substitutions (AURKA T287D, one residue from the site; PAK1 K299R, present in both states); a further pair outside the pure stratum (IKBKB) uses an S177E/S181E phosphomimetic non-phospho reference. See Sec. 3.2 and the confound audit shipped with the data.

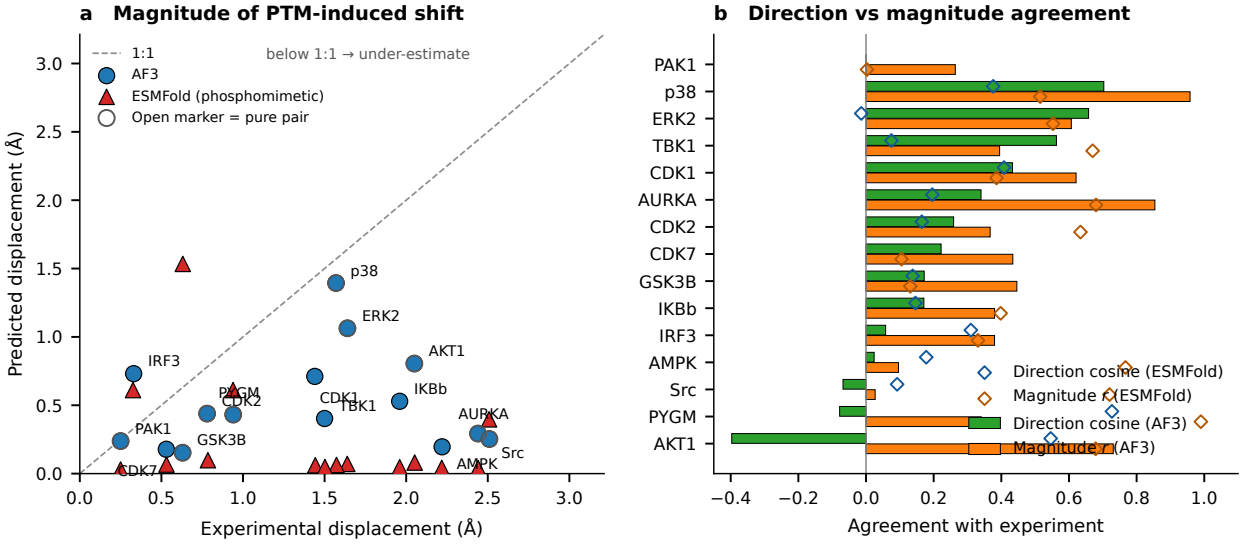


Figure 2: **Predicted versus experimental phosphorylation-induced displacement.** (a) Per-protein mean experimental displacement against the predicted displacement for AF3 (circles) and ESMFold with a phosphomimetic phospho state (triangles); the dashed line is 1:1. Open markers denote purely paired structures. (b) Per-protein direction cosine and magnitude correlation for AF3 (bars) and ESMFold (diamonds). Neither predictor's displacement tracks the experimental one.

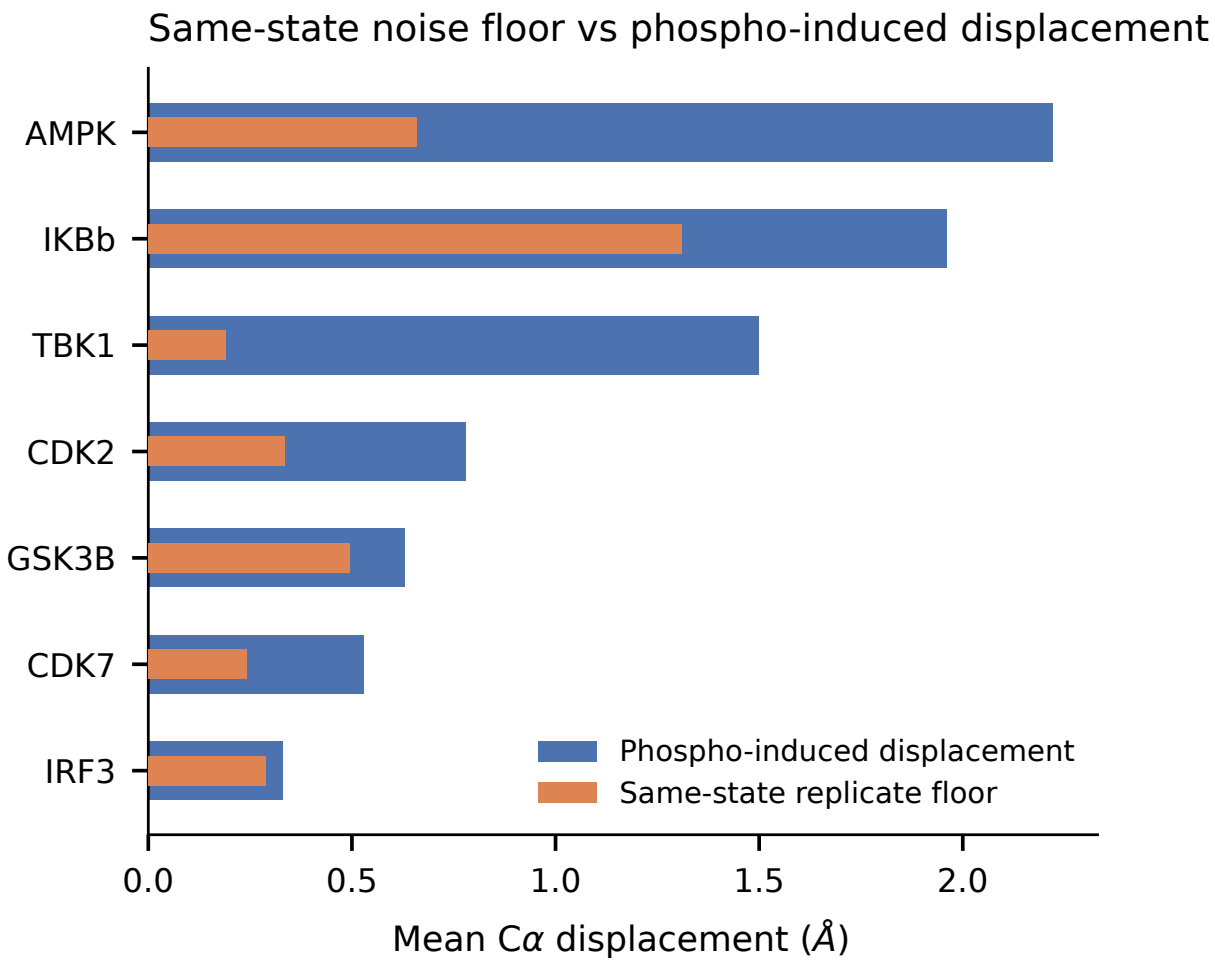


Figure 3: **Same-state noise floor.** Experimental same-state dispersion (crystallographically independent copies) and predictor same-state dispersion (across AF3 models) compared with the phosphorylation-induced experimental displacement.

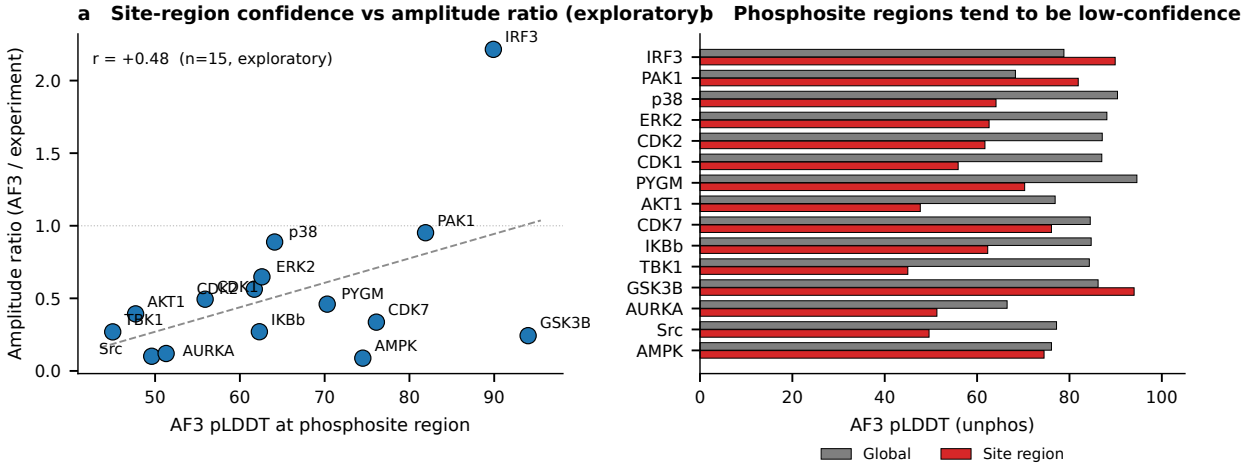


Figure 4: **Site pLDDT and agreement.** (a) Site-region pLDDT against amplitude ratio for the 15 pairs (labelled), with a least-squares fit; the apparent association is carried by IRF3 and collapses when it is removed (Sec. 4.5). (b) Site-region versus global pLDDT, showing that phosphosite regions tend to be lower-confidence than the rest of the chain.

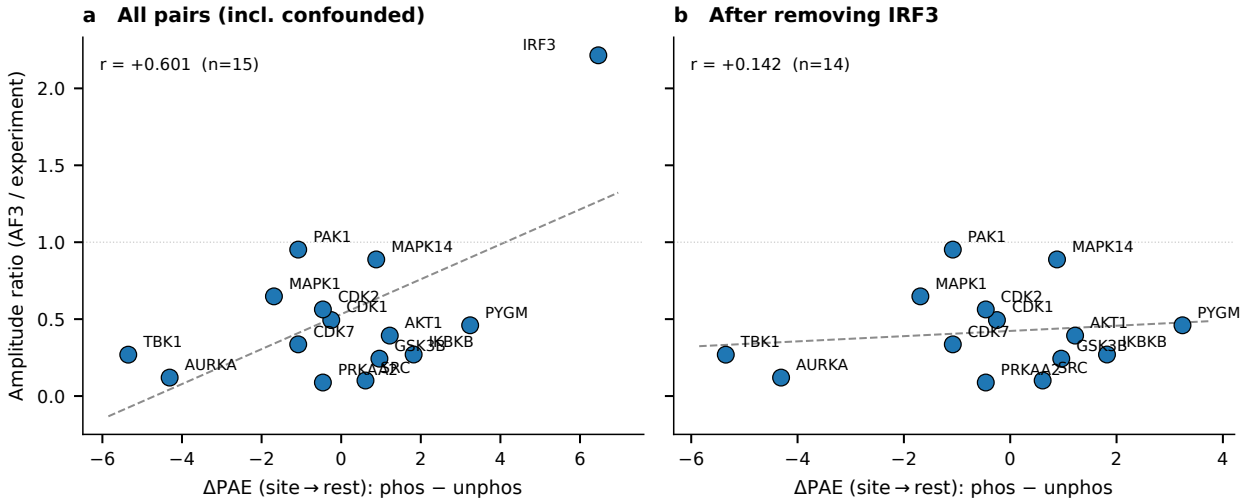


Figure 5: **ΔPAE association is a single-point effect.** ΔPAE (site→rest) against amplitude ratio, all pairs (left) and after removing IRF3 (right).

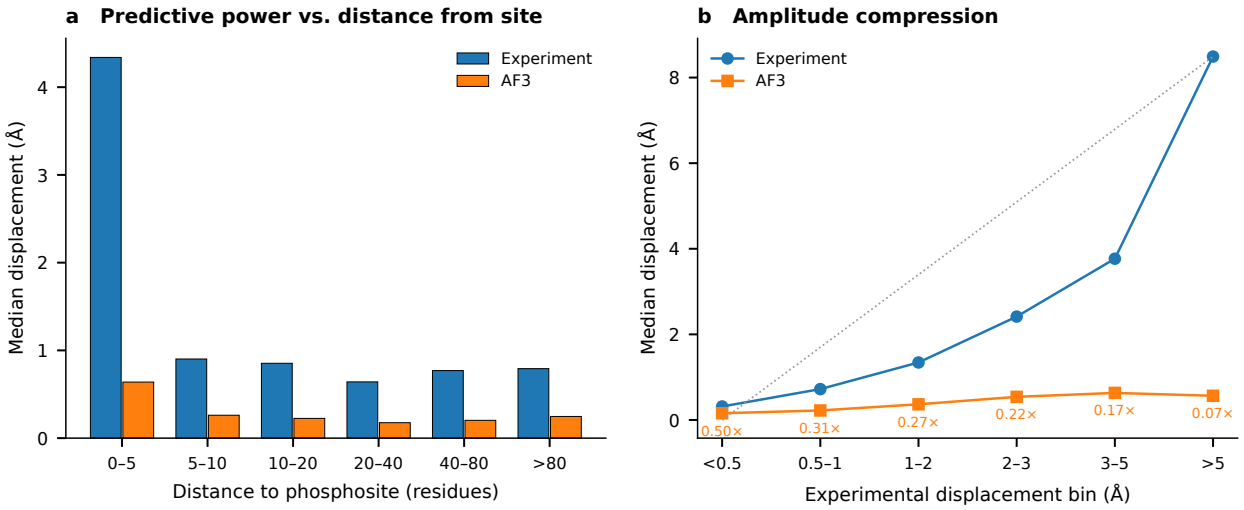


Figure 6: **Residue-level response saturates.** Median predicted/experimental displacement ratio as a function of experimental displacement magnitude, over all 5,148 common residues across the 15 cohort proteins; the decline is monotone across the six bins. The analysis uses the first AF3 model of each state (model 0), not a multi-model average. A distance decomposition of the same residues shows no trend (Sec. 4.6).

Supplementary material

Quantity	Value
n proteins	15
Pearson r (pred vs exp)	+0.1132
Pearson 95% CI	[-0.3433, +0.5744]
slope (pred per exp, Å/Å)	+0.0530
Spearman ρ	+0.1536
Spearman p	0.5848
mean predicted disp (Å)	0.521
SD predicted disp (Å)	0.359
experimental disp range (Å)	[0.25, 2.51]
max leverage (removed)	Src
r without max-leverage point	+0.2202
ESMFold Pearson r	-0.3622
ESMFold Pearson 95% CI	[-0.7175, +0.3079]
ESMFold mean predicted disp (Å)	0.253
ESMFold max leverage (removed)	PAK1
ESMFold r without max-leverage point	-0.4705
AF3 r excl. P3-failing pairs ($n = 13$)	+0.2134
AF3 r under direct superposition	+0.1214
AF3 mean predicted disp, direct (Å)	0.525
AF3 r, pure stratum ($n = 9$)	+0.2213 [-0.3895, +0.9109]
AF3 r, impure stratum ($n = 6$)	-0.2385 [-1.0000, +0.8369]
pure-vs-impure difference, permutation p	0.4007

Table S1: Cohort-level statistics for the AF3 (and, where available, ESMFold) predicted-versus-experimental displacement comparison. Bootstrap CIs resample proteins (the independent unit), not models.